

Lecture 2: Probability Toolkit

2.1 Random Variables & Associated Distributions

• Discrete r.v.

- home team wins $W \in \{0, 1\}$

• pmf:

$$P(W=1) = p, P(W=0) = 1-p$$

} Bernoulli

• Continuous r.v.

- margin of victory

• pdf:

$$P(a < M \leq b) = \int_a^b f(m) dm$$

• cdf:

$$F(x) = P(X \leq x)$$

* win probability & cover probability are CDF evaluations at different thresholds

2.2 Expectation & Variance

discrete

$$E[X] = \sum_x P(X=x) x$$

continuous

$$E[X] = \int x f(x) dx$$

• Expected value of a bet

- -120: risk \$100 to win \$83.33

- model probability 0.69

$$E[\text{profit}] = 0.69(83.33) + 0.41(-100) = \$8.17$$

* A positive value means the bet is worth taking if the model is correct

• Variance

$$\text{Var}[X] = E[X^2] - (E[X])^2$$

Bernoulli R.V. $\rightarrow \text{Var}[X] = p(1-p)$

$\rightarrow$ largest at $p=0.50$

$$\text{Var}[X+Y] = \text{Var}[X] + \text{Var}[Y] + 2\text{Cov}(X, Y)$$

$$\text{Cov}[X, Y] = E[(X - E[X])(Y - E[Y])]$$

correlation $\rho = \text{Cov}(X, Y) / (\sigma_X \sigma_Y) \rightarrow \text{rescales } [-1, 1]$

* 2008 Financial crisis: mortgage defaults were modeled as nearly independent when they were not

2.3 Conditional Probability

• Injury news

- starting QB is questionable

historically 70% play rate

- Friday fully practices

• 60% who end up playing Friday play

• 10% don't

Thurs → 70%

Sunday → 93%

$$P(\text{plays} | \text{Fri}) = \frac{0.6 \cdot 0.7}{0.6 \cdot 0.7 + 0.1 \cdot 0.3} = 0.93$$

2.4 Odds, Logs, & Logistic Function

The odds of an event with probability p : $\text{odds} = \frac{p}{1-p}$

Log odds stretch scale over whole real line: $\text{logit}(p) = \ln\left(\frac{p}{1-p}\right)$

↓
Inverse is the logistic function: $\sigma(x) = \frac{1}{1+e^{-x}}$

• Elo is a logistic model

$$E_{\text{home}} = \frac{1}{1+10^{-D/400}} = \sigma\left(\frac{\ln(10)}{400} D\right)$$